Μετάλλιον Inc.

POLYNESIAN POLYMETALLIC NODULES

Regional prospectivity and high-grade metal targets across the central and South Pacific

Exploration focus
Cook Islands • French Polynesia • Samoa Basin • Manihiki and adjacent oceanic plateaus

TECHNICAL GEOLOGY & EXPLORATION STRATEGY

Prepared for Μετάλλιον Inc.
Research cut-off: 2 November 2026

Assessment standard: evidence-led regional screening. Documented occurrences, resource estimates, geological analogues and untested hypotheses are kept separate.

EXECUTIVE DECISION

Where are the strongest Polynesian targets?

The best established loose-nodule province is the Cook Islands EEZ, especially the South Penrhyn Basin–Aitutaki Passage corridor for cobalt-rich nodules and the northern Penrhyn Basin for locally nickel- and copper-enriched nodules. For the highest cobalt concentration reported in the Polynesian region, the standout evidence is not an abyssal nodule field but the Tuamotu–Niau Fe-Mn crust and nodule-cluster province in French Polynesia. The Samoa Basin is now a high-priority emerging field because NOAA/USGS surveys in 2026 directly observed widespread nodules and released new abundance and geochemical datasets.

Key conclusion

For a project specifically targeting loose polymetallic nodules, prioritise the Cook Islands. For a high-grade cobalt thesis that includes Fe-Mn crusts and crust-hosted nodule clusters, add the southern Tuamotu / Niau sector. For a new, incompletely assayed nodule frontier, add the central and eastern Samoa Basin.

Decision summary

Priority	Area / target	Best metal thesis	Evidence status
1	Cook Islands: South Penrhyn Basin and Aitutaki Passage	Co-rich nodules; high nodule abundance; Fe, Co, Ni, Cu basket	High: regional surveys plus current resource estimates
2	Tuamotu / Niau, French Polynesia	Very high Co in Fe-Mn crust layers; Pt and Ni also reported; rare plateau-top nodule clusters	High-grade occurrence documented; no modern block resource
3	Samoa Basin / American Samoa OCS	Potential Co-Ni-Cu nodules; extensive fields observed in 2026	Occurrence confirmed; detailed assays / resource definition pending interpretation
4	North Penrhyn Basin	Local high-Ni / high-Cu nodule facies	High-grade samples reported; distribution and continuity require infill
5	Manihiki margins and other French Polynesian volcanic chains	Nodules on plateau / abyssal plain; Co-rich crust analogues on seamounts	Low to moderate: sparse or analogue-based in many sectors

Important distinction

Cobalt values of 0.7–1.3% reported for Tuamotu samples are bulk crust grades; values up to 2% refer to the outermost oxide layers in selected samples. They must not be presented as average grades of a free nodule field. Conversely, Cook Islands resource grades are nodule grades reported on a dry basis, while resource tonnage is reported as wet nodules. The two datasets are not directly interchangeable. [1–5]

What the report means by “higher grade”

The report ranks metal-grade opportunity by deposit type and metal: cobalt in Tuamotu crusts; cobalt plus nodule abundance in the southern Cook Islands; nickel and copper in the northern Penrhyn facies; and emerging, yet-to-be-ranked metal potential in the Samoa Basin. Manganese is important to the mass of the deposit, but Cook Islands nodule Mn grades are lower than typical Clarion-Clipperton Zone (CCZ) material.

01 | REGIONAL GEOLOGICAL FRAMEWORK

Polynesia is a family of contrasting seabed systems

The central and South Pacific contains both broad abyssal plains, where loose manganese nodules can accumulate at the sediment-water interface, and long-lived volcanic edifices, where cobalt-rich ferromanganese crusts precipitate directly onto exposed hard rock. The regional geomorphology includes the Manihiki Plateau, Penrhyn and Samoa-Niue basins, the Southern Cooks Basin, the Tuamotu and Society volcanic chains, and multiple seamount and fracture-zone systems. A 2026 regional map interpreted approximately 24.7 million km² of the southwestern Pacific Plate and identified multiple abyssal, plateau, volcanic-rise and seamount domains. [6]

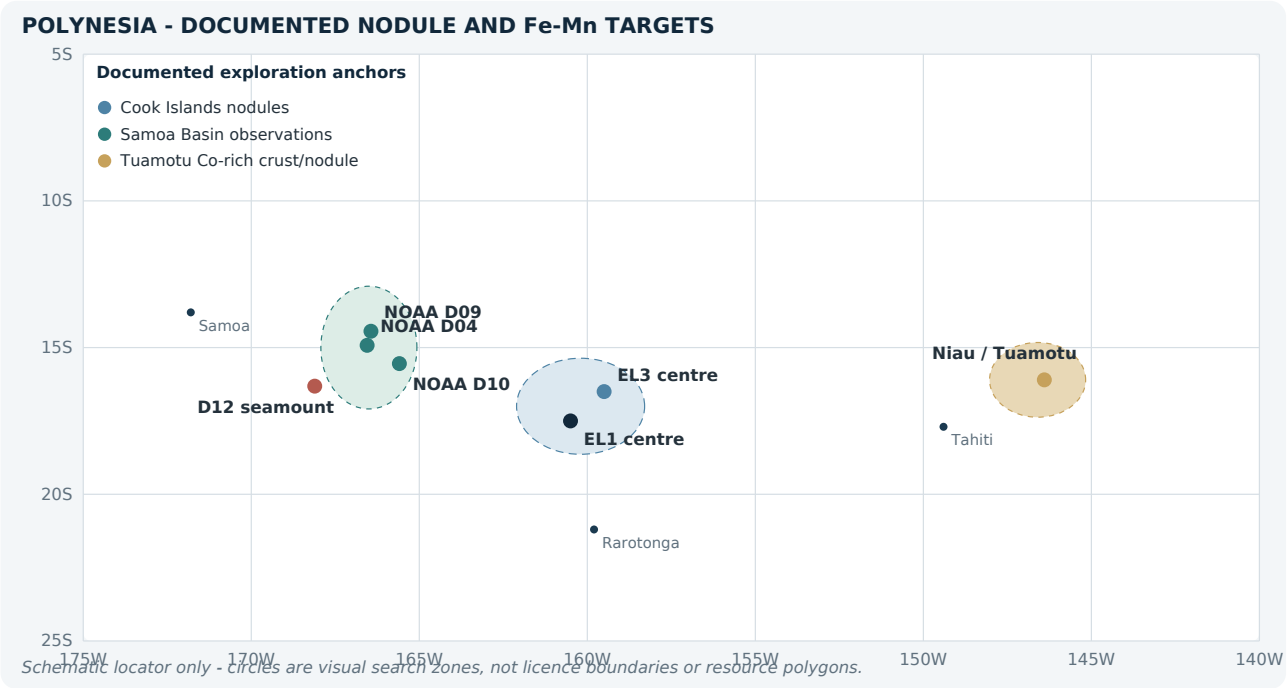


Figure 1. Schematic locator for the main evidence-backed targets. NOAA dive points are exact reported ROV start coordinates; EL1 / EL3 locations are project centres; Niau is a regional locator and not a sample coordinate.

Two deposit models drive the exploration strategy

Deposit model	Typical setting	Most prospective metals	Exploration implication
Abyssal polymetallic nodules	Flat, low-sedimentation abyssal plains, commonly 4,500–5,500 m; exposed or partly buried nodules at the sediment-water interface	Mn, Fe, Ni, Cu, Co; possible REY, Ti, Mo, Nb, V, W, Zr and Pt as trace / by-products	Map nodule abundance in kg/m², then assay bulk nodule chemistry and estimate grade continuity
Hydrogenetic Fe-Mn crusts and crust-hosted nodules	Hard-rock substrates on seamount flanks and plateaus, often hundreds to ~2,000 m in the Polynesian examples	Co, Mn, Fe, Ni, Cu, Pt, REY and other trace elements	Map substrate exposure, crust thickness and water-depth / oxygen-minimum-layer relationships; test cutting and recovery methods

The strongest high-grade opportunities are often spatially separated. High cobalt is favoured by hydrogenetic growth and well-oxygenated bottom water; higher Ni-Cu can occur where organic-matter export and diagenetic metal mobilisation are stronger. The Cook Islands technical report explicitly distinguishes high-Co southern facies from high-Ni northern facies. [2]

02 | METAL-GRADE BENCHMARK

What grades are actually supported by data?

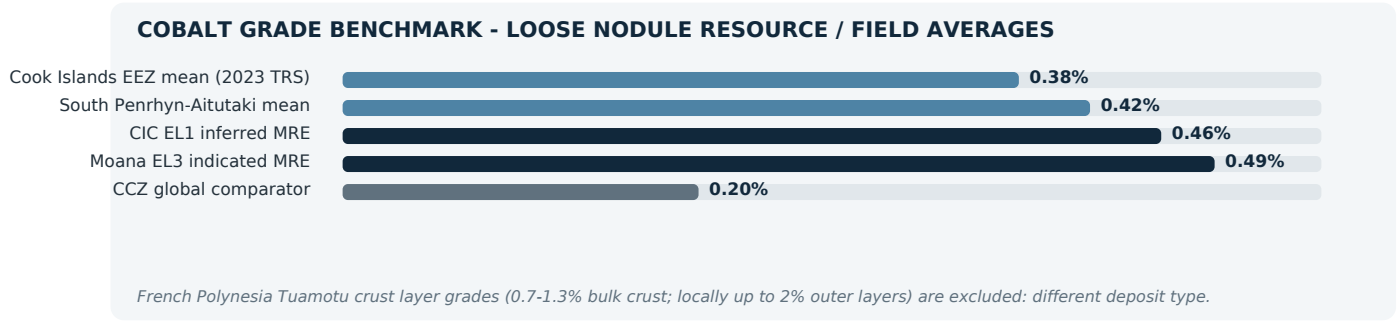


Figure 2. Cobalt grades in loose-nodule datasets. Values are published regional averages or reported mineral-resource estimates, not a uniform sample population. Cook Islands regional mean is from the 2023 technical-report table; CCZ comparator is from Hein et al. (2015); EL1 is an inferred resource; EL3 is an indicated resource. [1,2,3,4]

Dataset / sub-area	Co wt.%	Ni wt.%	Cu wt.%	Mn wt.%	Interpretation
Cook Islands EEZ mean (2023 TRS)	**0.38**	0.44	0.27	16.3	Co-enriched versus CCZ; moderate Ni-Cu mean
South Penrhyn Basin–Aitutaki Passage	**0.42**	0.37	0.21	16.2	Cobalt-led average; high abundance corridor
CIC EL1 inferred MRE	**0.46**	0.33	0.19	15.7	1,950 Mt wet; 19.9 kg/m² average abundance
Moana EL3 indicated MRE	**0.49**	0.27	0.15	15.6	417 Mt wet indicated resource; EL3 subset
Clarion-Clipperton Zone comparator	0.20	**1.20**	**0.98**	26.3	Stronger typical Ni-Cu-Mn mix, lower Co

High-grade sub-populations matter more than an average

The Cook Islands technical report describes a high-Co nodule class with Co above 0.26%, Fe above 13%, Mn below 20%, Ni below 0.40% and Cu below 0.30%; it is most common in the southern islands and western South Penrhyn Basin. A separate high-Ni class is concentrated in northern Penrhyn and adjacent areas, with Ni above 0.9% and locally up to 1.45%, and Cu above 0.6%. Thus, the best cobalt target and best nickel-copper target are not necessarily the same ground. [2]

Critical trace metals

Cook Islands nodules analysed by Hein et al. averaged approximately 1.20% Ti and 0.167% total rare-earth elements plus yttrium (REY); the study also identified high contents of Mo, Nb, V, W and Zr, and enrichment relative to the CCZ in Te, Pt and other high-tech / energy metals. These are strategic by-product signals, not yet independently demonstrated payable products. The report recommends full multi-element assay and metallurgical deportment testing before assigning economic value. [1]

03 | COOK ISLANDS

The benchmark loose-nodule province

The Cook Islands EEZ contains extensive nodule-bearing abyssal plains in the Penrhyn, Samoa-Niue and Southern Cooks basins. A long history of regional sampling, four JICA/MMAJ expeditions between 1985 and 2000, modern exploration and government-commissioned resource work gives this area the strongest exploration dataset in Polynesia. The Cook Islands seabed minerals authority describes nodules as widespread, while the RSC technical report explains that the best-quality nodule plains are associated with appropriate sedimentation, nucleus supply and bottom-current conditions. [1,2]

Priority sub-targets

Target	Evidence / metal thesis	Recommended test
South Penrhyn Basin–Aitutaki Passage	Highest regional focus for abundant Co-rich nodules. Mean grades in this sub-area: ~0.42% Co, 0.37% Ni, 0.21% Cu and 16.2% Mn. High-Co class is linked in part to oxygenated bottom currents. [1,2]	Infill box cores and AUV/ROV imagery across abundance highs; collect whole-nodule and layer-specific multi-element assays.
Northern Penrhyn Basin	High-Ni subgroup is reported here: >0.9% Ni and up to 1.45% Ni, with Cu >0.6%; adjacent parts of Kiribati are also described. [2]	Separate high-Ni domains from high-Co domains; verify Cu, Ni, Co continuity and density on a tighter grid.
EL1 / north-central Cook Islands project area	EL1 resource centre approx. 17.5°S, 160.5°W. CIC reports 1,950 Mt wet nodules, 19.9 kg/m² mean abundance, at 13 kg/m² cut-off; grades 0.46% Co, 0.33% Ni, 0.19% Cu, 15.7% Mn and 17.8% Fe (dry-basis grades). Inferred only. [3]	Validate untested licence blocks, improve bathymetry and sample spacing, and confirm moisture / dry-tonnage factors.
EL3 / central Cook Islands project area	EL3 centre approx. 16.5°S, 159.5°W. 2026 filing reports 417 Mt wet indicated and 102 Mt wet inferred; indicated grades include ~0.49% Co, 0.27% Ni, 0.15% Cu, 15.6% Mn and 18.9% Fe. [4]	Treat as a benchmark sub-project; avoid double counting with wider EEZ inventory. Check resource boundary and effective date.

What the resource numbers do - and do not - say

The CIC EL1 estimate is classified entirely as Inferred; the RSC report states that no Measured or Indicated material was estimated, no Mineral Reserve was declared and no economic viability was demonstrated. Its grades are reported on a dry basis while the 1,950 Mt tonnage is wet; the report estimates contained metals using a 25% average moisture factor. The resource is an in-situ estimate before collection and processing, and metallurgical recovery has not been applied. The EL3 estimate is a separate licence-area estimate and should not be added to the full EEZ legacy inventory as though the datasets were independent. [3,4]

Cook Islands conclusion

For loose nodules, this is the first place to spend exploration capital. Choose southern Penrhyn / Aitutaki Passage for cobalt and high abundance; choose northern Penrhyn for a nickel-copper-led target. Do not optimise all three metals on a single average grade.

04 | FRENCH POLYNESIA

Tuamotu–Niau: the standout cobalt anomaly

French Polynesia contains the most compelling published high-cobalt Fe-Mn occurrence in the Polynesian region. The NODCO I survey (1986) examined a restricted sector near Niau Island on the southern flank of the Tuamotu Archipelago. The site is a drowned / subsided fossil atoll plateau at approximately 1,000 m water depth. Fe-Mn crust outcrops occur on hard substrate around the former lagoon rim; rare nodules and slabs are associated with the crusts, and nodule clusters have been described on plateau tops. [5,6]

Reported parameter	Published result	Interpretation for Μετάλλιον Inc.
Crust thickness	Average about 2.5 cm on seamount plateaus / slopes in the 1987 study.	A thin, surface-bound deposit; thickness, substrate exposure and continuity are key tonnage controls.
Cobalt - Niau zone	Bulk crust Co reported at approximately 0.7–1.3%; the thinnest crusts on plateau flanks carry the highest values. [6]	High-grade Co is real, but it occurs in crust samples rather than a defined loose-nodule resource.
Cobalt - oxide layers	A related 1987 study found Co increasing from about 0.5% in inner oxide layers to up to 2% in peripheral layers. [5]	Layer-specific peak is not equivalent to whole-deposit average. Requires systematic vertical / lateral sampling.
Nickel / copper / platinum	Average Ni about 0.5%, Cu about 0.1%; Pt about 0.2–1.3 ppm in the Niau crust study. [6]	Cobalt is the primary signal; Ni is secondary, Cu modest, Pt may be a trace by-product.

The geological control

The Niau deposits consist of ferromanganese oxide zones associated with a phosphatised reefal / carbonate substrate and underlying volcanic framework. Authors relate cobalt enrichment to the oxygen-minimum layer and to fine, poorly crystallised δ-MnO_2. The highest reported cobalt values occur in the thinnest crusts on plateau flanks rather than necessarily in the thickest crust. This is a critical exploration warning: grade and thickness may trade off. [5,6]

Regional extension beyond Niau

The Tuamotu, Society, Tarava and Austral chains comprise numerous extinct volcanic edifices, rises and seamounts. A 2026 geomorphological synthesis identifies Fe-Mn crusts as the only currently known seabed mineral type associated with these long-extinct volcanic edifices. This supports a regional crust prospectivity thesis, but it is not evidence that every seamount carries thick, cobalt-rich crust or loose nodules. [7]

Tuamotu conclusion

If the objective is highest cobalt grade rather than the largest known loose-nodule resource, the Niau / southern Tuamotu crust province is the most attractive Polynesian lead. Begin with a data-recovery and modern verification programme; do not book a 0.7–2.0% Co grade as an exploitable bulk resource.

05 | SAMOA BASIN / AMERICAN SAMOA

A newly verified nodule frontier

The Samoa Basin is now a compelling, rapidly improving exploration target. In April 2026, USGS scientists deployed 38 box cores over 13 April–28 April; 36 recovered enough material to determine nodule density and 35 supported subcore collection. The released dataset includes station coordinates, depths, abundance in kg/m², sample type and representative geochemistry. The surveyed area spans approximately 13.17–17.04°S and 167.86–165.62°W. [8,9]

NOAA ROV observations provide direct visual anchors

Dive / field	WGS84 start coordinate	Depth	Observation
Dive 04 - East Samoa Basin	14.92326°S, 166.55247°W	~5,140 m	Medium-density abyssal field; nodule density increased over a small mound. [10]
Dive 09 - Central Samoa Basin	14.44314°S, 166.43847°W	~5,508–5,512 m	Relatively homogeneous nodule distribution; less than 50% visible bottom cover. [11]
Dive 10 - East Samoa Basin	15.54622°S, 165.59267°W	~5,524–5,529 m	Highest observed nodule abundance of the expedition at that point; approximately 80% of seafloor covered during the dive. [12]
Dive 12 - South Samoa seamount	16.31471°S, 168.11167°W	~3,070–3,262 m	Seamount slope: Fe-Mn-encrusted outcrops, boulders and cobbles interspersed with loose nodules and sediment. [13]

Interpretation

The 80% visual cover at Dive 10 is a strong nodule-field indicator, but it is not a calibrated kg/m² abundance estimate, a grade assay or a resource classification. NOAA/USGS have released the underpinning abundance and representative-geochemistry datasets, making this a practical next target for data integration. The geochemical dataset contains 120 point records; the separate abundance dataset contains 38 box-core points. [8]

A major near-term jurisdictional milestone

The US Federal Register published the final notice for American Samoa OCS Pacific Mineral Lease Sale 1 on 20 October 2026, scheduling an auction for 19 November 2026 and offering two large offshore lease areas. The leases cover minerals in polymetallic nodules and ferromanganese crusts. A winning lease would not itself authorise immediate mining or harvesting; later regulatory approvals and plans are still required. This is a current licensing process in US federal waters, not a statement about the status of American Samoa territorial waters. [14,15]

Samoa conclusion

Treat the Samoa Basin as a high-priority data-acquisition target with strong field confirmation and new public sample datasets, but an unranked grade profile until Co-Ni-Cu and trace-element assays are interrogated sample by sample.

06 | SECONDARY PROVINCES

Where else should Μετάλλιον look?

The broader Polynesian region contains multiple volcanic chains and abyssal basins, but the evidence becomes weaker away from the Cook Islands, Tuamotu and Samoa Basin. These are screening leads rather than resource opportunities.

Area	Geological case	Likely target type / metals	Current ranking
Manihiki Plateau and eastern / southern margins	Ancient oceanic plateau with faulted margins, horsts, internal rifts and adjacent abyssal plains. Nodules occur on abyssal plains and have been reported on the plateau, but the plateau is poorly sampled. Fe-Mn crusts occur around the plateau. [2]	Nodule patches on suitable deep, flat surfaces; Co-rich crusts on exposed hard substrate. Co, Ni, Cu, REY and trace metals to test.	Moderate geological potential; low confidence in grade / tonnage
Society, Tarava and Austral chains	Extinct volcanic edifices and seamounts are abundant; the 2026 regional map identifies this as a crust-type geological domain. [7]	Fe-Mn crusts and trace-metal enrichment; no basis yet to assume extensive loose nodule plains on steep edifices.	Moderate crust analogue; low nodule evidence
Pukapuka Ridge / Tuamotu Plateau transition	Complex volcanic ridge and seamount terrain; geomorphological and tectonic maps improve targeting, but a continuous high-grade nodule resource is not demonstrated. [7]	Crusts on hard surfaces; nodules only where wide, sedimented, low-gradient deep plains occur.	Reconnaissance only
Niue / Tokelau / Pitcairn / international-water corridors	Abyssal plains and seamounts create permissive geological conditions; regional datasets are sparse. Do not infer a nodule field from bathymetry alone.	Possible nodule and crust occurrences; metal suite unverified.	Low until direct sampling
Tonga-Lau and trench / arc margin	Strong tectonic and volcanic relief, but steep slopes and trench settings are less favourable for broad, surface nodule blankets. Hydrothermal mineral systems are a separate model.	Fe-Mn coatings / crusts on seamounts; possible hydrothermal sulphides in distinct tectonic settings.	Low for large loose-nodule fields

Regional screening rule

Prioritise large, low-gradient abyssal plains below the carbonate compensation depth for loose nodules. Use seamount chains, submerged atolls and volcanic plateaus to target Fe-Mn crusts. Avoid treating a steep seamount, ridge crest or trench flank as a nodule target merely because ferromanganese coatings are visible. [2,7]

07 | EXPLORATION RANKING

Ranked targets by the metal thesis

Target rank	Target	Best metal	Evidence score	Main reason / key uncertainty
1 - Nodule	South Penrhyn Basin–Aitutaki Passage, Cook Islands	Co + Ni + Cu basket	**Very high**	Strong abundance and high-Co facies; need block-scale re-assay, current density maps and environmental baselines.
2 - High-grade Co	Niau / southern Tuamotu, French Polynesia	Co; Pt / Ni trace by-products	**High occurrence / low resource confidence**	0.7–1.3% bulk crust and up to 2% outer layers reported; crusts dominate, nodules are rare and clustered. No modern resource model.
3 - Emerging nodules	East / central Samoa Basin	Co-Ni-Cu (unranked)	**High occurrence / assays pending review**	ROV verified; ~80% visual cover at Dive 10 and 2026 box-core datasets; published grade ranking not yet established.
4 - Ni-Cu	Northern Penrhyn Basin	Ni + Cu	**High samples / continuity uncertain**	Ni >0.9% to 1.45% and Cu >0.6% in high-Ni facies; grade may trade off with Co.
5 - Crust extension	Other Tuamotu / Society / Austral seamounts	Co + REY + Pt (to assay)	**Moderate analogue**	Extinct volcanic edifices are permissive for crusts; grade and thickness are not demonstrated outside known sites.
6 - Frontier	Manihiki Plateau margins / Niue-Tokelau-Pitcairn	Unknown	**Low to moderate**	Large gaps in sampling; basin geometry and depth need to be matched to nodule-forming conditions.

Separate rankings by commodity

Commodity objective	First choice	Second choice	Rationale
Cobalt - loose nodules	South Penrhyn / Aitutaki Passage	EL1 / EL3 Cook Islands areas	Cook Island mean / MRE grades around 0.38–0.49% Co and strong abundance.
Cobalt - highest local grade evidence	Tuamotu / Niau crust province	Cook Islands high-Co nodule belt	Tuamotu bulk crust ~0.7–1.3% Co; up to 2% in outer oxide layers. Different deposit type.
Nickel + copper	Northern Penrhyn high-Ni facies	Samoa Basin - pending assays	Cook Islands high-Ni samples >0.9% Ni, locally to 1.45%, and Cu >0.6%.
REY / Ti / W / Nb / V / Zr	Cook Islands nodule samples	Tuamotu Fe-Mn crusts - analytical extension	Published Cook Islands trace-element enrichment; regional crust potential needs modern multi-element assay.

08 | RECOMMENDED EXPLORATION PROGRAMME

A staged programme to de-risk grade and continuity

The immediate opportunity is to extract more value from existing data, not to commission a full deep-ocean campaign before the public and historic datasets have been consolidated. A disciplined, two-stream programme should keep loose-nodule and crust targets separate.

Phase / timing	Workstream	Key deliverables	Decision gate
0. Data room **0–3 months**	Acquire and harmonise JICA-MMAJ, SBMA, CIC/Moana, USGS, NOAA, PANGAEA and published French Polynesia NODCO datasets. Reproject to WGS84; validate sample coordinates, depths, sample type, dry/wet basis and QA/QC.	Georeferenced target database; sample-level chemistry matrix; uncertainty ledger; map of sampled vs unsampled seafloor.	Do not deploy until data gaps are visible and assays are traceable.
1. Reconnaissance **3–9 months**	Cook Islands: re-sample high-Co and high-Ni facies. Samoa: integrate 2026 box cores and ROV sites. Tuamotu: recover NODCO I sample-level data and verify precise sample locations.	First-pass prospectivity surfaces; confirmatory assays; candidate 1,000–10,000 km² search areas.	Advance only targets with repeatable grade plus abundance / thickness evidence.
2. Focused surveys **9–18 months**	AUV high-resolution imagery, multibeam, backscatter, sub-bottom profiling, ROV transects, box cores and dredges where appropriate. Use separate tools for abyssal nodules and hard-substrate crusts.	Nodule abundance grids (kg/m²); crust thickness grids (cm); slope / roughness model; sample-level spatial grade model.	Reject areas with poor grade continuity, steep slopes, sediment cover or unacceptable ecological constraints.
3. Metallurgy & baseline **12–24 months**	Bulk composite tests by facies; mineralogical deportment (XRD, SEM-EDS, EPMA, LA-ICP-MS); hydrometallurgical leach; residue and seawater-impact testing. Baseline benthic and water-column ecology.	Recovery curves and reagent balance; product purity; residue characterisation; baseline environmental data.	No commercial case without technically credible recoveries, waste controls and consent pathway.
4. Resource definition **18–36 months**	Infill sampling and independent competent-person review under a recognised reporting code. Estimate nodule resources or crust inventory separately; do not combine them.	Resource model with grade, tonnage / area, confidence classes, recovery assumptions and environmental exclusion zones.	Only then progress to scoping economics and pilot-collection design.

Priority sample suite

For every loose nodule: whole-nodule assay (Co, Ni, Cu, Mn, Fe, Ti, Zn, Mo, V, W, Nb, Zr, REY+Y, Pt, Te, Ba, P, Si, Al, Ca), moisture, dry density, size distribution, nucleus type and layer-specific mineralogy. For each crust: separate outer, middle and basal layers; thickness; substrate lithology; porosity; fracture density; continuous outcrop mapping; trace-element profiles. In both cases, retain standards, blanks, duplicates and chain-of-custody records.

09 | PROCESSING & COMMERCIAL FILTER

High assay does not equal recoverable metal

Polymetallic nodules are fine-grained mixtures of manganese oxide and iron oxyhydroxide phases that host multiple metals. The metals can be chemically bound to different mineral phases; a strong total assay does not guarantee a high recovery, clean product or attractive payable terms. The Cook Islands EL1 technical report notes that no metallurgical recovery estimate has yet been applied to the resource, and no full economic or cash-flow analysis was completed. [3]

Processing questions to resolve

Question	Why it matters	Test work required
Cobalt, nickel and copper deportment	Determine which oxide phases / growth layers host payable metals and how variable they are between facies.	XRD, SEM-EDS, EPMA, MLA/QEMSCAN where appropriate; sequential leaching and solution chemistry.
Manganese and iron load	High Mn/Fe mass can add reagent, separation and residue demands; iron may be a non-pay metal.	Bench-scale hydrometallurgy; acid consumption; impurity removal; Mn product route assessment.
Trace-metal by-products	REY, Ti, W, Nb, V, Zr, Te and Pt may materially affect value only if they can be selectively recovered to saleable quality.	Element-specific deportment, concentration and recovery tests; marketable product specification and offtake review.
Wet mass and dry-grade basis	A mismatch between wet resource tonnage and dry assay grade can overstate contained metal if moisture is not measured reliably.	On-deck wet weighing, full-sample drying, duplicates and in-situ moisture / density checks.
Crust mining vs nodule collection	Nodules are loose surface clasts; crusts are bonded to hard substrate and may need cutting or mechanical removal.	Separate collector prototypes, dilution / substrate recovery tests, cutting energy, product contamination and recovery balance.

Commercial screening framework

A viable target must pass five independent filters: (1) grade and abundance / thickness; (2) spatial continuity; (3) recovery and concentrate quality; (4) collection, pumping, transport and processing cost; and (5) environmental and social licence. A high-Co crust at 1,000 m may be easier to reach than a nodule field at 5,500 m, but crusts are more difficult to detach cleanly from substrate. Conversely, abyssal nodules can be collected as surface material, but the extreme depth increases vessel, riser and system complexity.

Commercial caution

Avoid valuing Ti, REY, W, Nb, V, Zr, Te or Pt at contained-metal spot value. No revenue should be booked until a recoverable stream, product specification, payable terms and process costs have been demonstrated.

10 | REGULATORY & ENVIRONMENTAL SCREEN

Jurisdiction and ecological risk are project fundamentals

The prospectivity map crosses several legal regimes. Cook Islands national waters are governed by the Seabed Minerals Act and supporting regulations; French Polynesia is an overseas collectivity within the French Republic, so any project must establish the applicable French and territorial competencies; and the new Samoa Basin field data include US federal waters beyond American Samoa territorial waters. Areas beyond national jurisdiction are subject to the international seabed regime under UNCLOS and the International Seabed Authority. Legal title, mineral rights and access must be verified separately from geological prospectivity.

Current jurisdictional points as at 2 November 2026

Jurisdiction	Current position / relevant point	Implication
Cook Islands	The SBMA lists the Seabed Minerals Act 2019 framework and regulations for exploration (2020), environmental matters (2023), harvesting and other mining (2024), and royalties. The EL1 technical assessment states that no commercial harvesting or mining had been authorised or commenced as of its effective report date. [3,16]	Exploration licences are not mining approvals. Environmental assessment, EIS, management plans and further approvals remain critical.
French Polynesia / France	Niau / Tuamotu occurrences sit within French Polynesian waters. The historic studies do not confer mineral title or current exploration rights. The specific resource-ownership and permission route must be verified with French and territorial authorities.	Treat historic grades as scientific evidence only; no licence or commercial right is implied.
American Samoa OCS	Final notice published 20 October 2026 for PACM-1 auction scheduled 19 November 2026; two OCS lease areas. The final notice expressly says later approvals are needed for development. [14,15]	Near-term regulatory event; lease auction is not an operating or extraction permit. Track community consultation and environmental stipulations.
High seas	Activities in areas beyond national jurisdiction are subject to the International Seabed Authority framework and UNCLOS.	No project assumption should be made about rights, exploitation rules or access without current legal review.

Environmental baseline priorities

Nodule fields and crust-covered seamounts are distinct deep-sea habitats. Baseline work should document benthic community composition, sediment grain size and geochemistry, megafauna, microbial communities, water-column particulates, currents, turbidity, noise and potential collector / plume effects. In the Samoa Basin the 2026 dives directly documented sponges, corals, fish and invertebrates among nodule habitat. [10–13] The high cobalt grade in Niau does not override the need to characterise reefal substrate, crust communities and ecological connectivity.

Precautionary principle

A resource estimate is not a social licence. The practical exploration strategy should use non-invasive imagery and mapping first, minimise sample mass, avoid sensitive habitats, publish baseline datasets where appropriate, and maintain a no-go framework for areas with high biodiversity or uncertainty.

11 | FINAL RECOMMENDATIONS

Recommended position for Μετάλλιον Inc.

A focused Polynesian strategy should pursue two separate deposit families: (A) loose nodules in abyssal basins, with the Cook Islands as the benchmark and Samoa as the emerging frontier; and (B) cobalt-rich crusts and crust-hosted nodules on volcanic edifices, with the Tuamotu–Niau sector as the best documented lead. Combining them in one regional “nodule grade” number would obscure the most important geological and engineering distinctions.

Five immediate actions

Action	Recommendation	Success measure
1. Build a verified regional data room	Compile sample-level data for Cook Islands, Samoa Basin and French Polynesia. Preserve original analyte units, depth, sample type, wet/dry basis and lab methods.	100% of priority records traceable to a source file / station ID; coordinate QA complete.
2. Secure Niau / Tuamotu primary data	Acquire the NODCO I cruise reports, dredge logs, sample locations and PANGAEA data; reconstruct exact survey lines and identify crust vs nodule sample types.	Georeferenced map of sample points and layer-by-layer Co-Ni-Cu-Pt data.
3. Reprocess Samoa Basin 2026 releases	Integrate the USGS abundance CSV and 120-point geochemistry release with NOAA Dive 04, 09, 10 and 12 imagery. Rank by density, depth, grade and ecological context.	Validated abundance / grade map; top three follow-up blocks.
4. Separate cobalt and Ni-Cu nodule plays	Use South Penrhyn / Aitutaki for Co-led abundance; northern Penrhyn for Ni-Cu; test correlations before defining a blended product.	Independent block-scale models for Co-led and Ni-Cu-led facies.
5. Keep crust and nodule economics separate	Commission distinct conceptual collector systems and metallurgical work for loose nodules and hard-substrate crusts. Do not extrapolate nodule resource cut-offs to crust.	Separate technical and commercial decision memos; credible recovery and environmental pathways.

Overall verdict

Best nodule prospect in Polynesia

South Penrhyn Basin–Aitutaki Passage, Cook Islands: the strongest combination of documented nodule density, cobalt-rich chemistry and resource definition. For nickel-copper, test the northern Penrhyn Basin separately.

Best high-grade cobalt prospect in Polynesia

Tuamotu–Niau, French Polynesia: published crust grades of about 0.7–1.3% Co, with local outer-layer values up to 2%. This is principally a crust / crust-hosted nodule target, not a defined abyssal loose-nodule deposit.

Best emerging new nodule frontier

Samoa Basin near American Samoa: 2026 ROV field observations and USGS box-core datasets are strong evidence of nodule occurrence; grade ranking remains open until the released assays are modelled.

Research limitation: this is a regional geological prospectivity assessment based on publicly available information as of 2 November 2026. It is not a mineral resource estimate, valuation, legal opinion or recommendation to undertake commercial seabed mining.

12 | SOURCES & TECHNICAL REFERENCES

Source register

The following primary and institutional sources support the grades, locations, resource statements and current regulatory context used in this report. Bibliographic links are included for traceability.

[1] Hein, J.R., Spinardi, F., Okamoto, N., Mizell, K., Thorburn, D. & Tawake, A. (2015). “Critical metals in manganese nodules from the Cook Islands EEZ, abundances and distributions.” Ore Geology Reviews. DOI: 10.1016/j.oregeorev.2014.12.011. USGS Publications Warehouse record. [Source link]

[2] Tay, S., Browne, R. & Bertoli, O. (2023). Cook Islands Polymetallic Nodule Deposit: Technical Report and Supporting Documentation for the Cook Islands Polymetallic Nodules Resource Estimate. Prepared for Cook Islands Seabed Minerals Authority; RSC Consulting. [Source link]

[3] RSC Consulting / CIC Limited (2026). Initial Assessment and Technical Report Summary for the EL1 Cook Islands Polymetallic Nodule Deposit. Effective 31 December 2025; revised 3 September 2026. SEC Exhibit 96.1. [Source link]

[4] RSC Consulting / Moana Minerals Ltd (2026). Mineral Resource Estimate for the EL3 Cook Islands Polymetallic Nodule Deposit. SEC Exhibit 96.2; resource tables and technical disclosure. [Source link]

[5] (1987). “Characteristics of Co-rich ferromanganese nodules and crusts sampled in French Polynesia.” Marine Geology, 77(1–2), 109–119. DOI: 10.1016/0025-3227(87)90086-7. [Source link]

[6] Le Suave, R. et al. (1989). “Geological and mineralogical study of Co-rich ferromanganese crusts from a submerged atoll in the Tuamotu Archipelago (French Polynesia).” Marine Geology, 87(2–4), 227–247. DOI: 10.1016/0025-3227(89)90063-7. Associated data: PANGAEA 855093. [Source link]

[7] Parianos, J., Browne, R., Uan, J. & Paul, C. (2026). “A regional seabed geomorphological map over part of the southwestern Pacific Plate.” Journal of Maps. DOI: 10.1080/17445647.2026.2627733. [Source link]

[8] Adamczyk, K.B. et al. (2026). Abundance and geochemistry of abyssal ferromanganese nodules and ferromanganese coated rocks from the Samoa Basin (2026-604-FA). USGS Data Release. DOI: 10.5066/P149BW9Z. [Source link]

[9] USGS (17 June 2026). “USGS Releases Expedition Report for Samoa Basin Box Coring Effort.” 38 box cores deployed; preliminary abundance and geochemistry programme. [Source link]

[10] NOAA Ocean Exploration (31 August 2026). Dive 04: East Samoa Basin Abyssal Nodule Field. Coordinates: 14.92325896°S, 166.55247106°W. [Source link]

[11] NOAA Ocean Exploration (8 September 2026). Dive 09: Central Samoa Basin Nodule Field. Coordinates: 14.44314197°S, 166.438469°W. [Source link]

[12] NOAA Ocean Exploration (9 September 2026). Dive 10: East Samoa Basin Nodule Field (12C). Coordinates: 15.54622024°S, 165.5926715°W. [Source link]

[13] NOAA Ocean Exploration (11 September 2026). Dive 12: South Samoa Basin Seamount. Coordinates: 16.314713449°S, 168.111670596°W. [Source link]

[14] Federal Register / U.S. Department of the Interior (20 October 2026). American Samoa Outer Continental Shelf Pacific Mineral Lease Sale 1, 91 FR 66352; Docket BOEM-2026-0331. [Source link]

[15] Marine Minerals Administration (19 October 2026). Final Leasing Notice for American Samoa OCS Pacific Mineral Lease Sale 1; explains lease sale and subsequent approvals. [Source link]

[16] Cook Islands Seabed Minerals Authority. Regulations under the Seabed Minerals Act 2019; current list includes Exploration Regulations 2020, Environment Regulations 2023, and Minerals Harvesting and Other Mining Regulations 2024. [Source link]

Method note

The report uses a hierarchy of confidence: (i) primary geochemical papers and official resource reports; (ii) official USGS / NOAA survey releases and ROV logs; (iii) regional geological mapping; and (iv) interpretive regional screening. Coordinates are shown only when an official report provides them or when they are clearly labelled as approximate locator positions. Grade comparisons preserve deposit type and dry / wet basis where stated. No new resource tonnage has been estimated for French Polynesia or Samoa Basin.

Prepared for Μετάλλιον Inc. | 2 November 2026